

Utah FORGE downhole microearthquake codas require small-scale 3-D heterogeneity: elastic tests of smooth, layered, fracture-zone and stochastic media

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Abstract

Downhole records of microearthquakes (MEQs) induced by the April 2022 stimulation at Utah FORGE contain long S-wave codas. We test which elastic media reproduce them. 416 MEQs recorded at 4 kHz by two three-component sensors in well 56-32 are compared with 3-D finite-difference wavefields computed by reciprocity for smooth media (homogeneous, 50-m sonic-log layering, 3-D gradients, transverse isotropy), deterministic small-scale media (6-m sonic-log layering, 5-m-thick fracture zones) and lognormal stochastic media with power-law, exponential and Gaussian spectra. Moment tensors fitted to the direct waves (P – 10 ms to S + 15 ms) predict the 40–80 Hz energy from S + 40 to S + 100 ms. The smooth media under-predict this late energy by median factors of 44–82 at the upper and 12–15 at the lower sensor. This deficit persists for alternative windows, deviatoric and double-couple sources, finite source duration, well-timed events and a worst-case in-band tool resonance; the simulated coda is converged in grid and absorbing boundary. Thin layers and fracture zones reduce it to factors of 11–19 (upper) and 2.4–3.4 (lower sensor), and stochastic media match it. The observed coda decorrelates between neighbouring events (0.69 to 0.49 from 0–5 to 80–160 m), whereas smooth and deterministic small-scale media keep it coherent (≥ 0.79) and

stochastic media decorrelate it. Within the isotropic lognormal model class, the late energy implies an effective scattering strength $\sigma_{\text{eff}} \approx 0.05\text{--}0.13$ (standard deviation of $\ln V$), depending on sensor and spectral family. A larger simulation domain raises the stochastic coda, so these values are upper bounds, comparable to the sonic-log value of 0.045. At 20–80 Hz neither per-event nor common-grid statistical comparisons (event bootstrap, holdout) distinguish power-law from exponential or Gaussian spectra, and the sonic-log spectrum ($\beta = 0.85 \pm 0.07$ over 12–200 m) does not exclude a short correlation length. Within elastic modelling, the late S energy thus requires strong small-scale 3-D heterogeneity, unless an unmodelled site response of comparable size is present; its spectral form is not resolved. Deterministic comparison against an unknown stochastic realisation favours smooth media, so medium-class selection should use wavefield statistics.

Keywords: EGS; Utah FORGE; microseismicity; seismic scattering; coda; random media; 3-D finite difference; downhole seismics

Introduction

Microseismic monitoring of engineered geothermal system (EGS) stimulation relies on a velocity model. In routine practice that model is homogeneous, layered or weakly anisotropic, smooth enough for ray theory to locate events and for a few source parameters to fit the direct waves. Everything after the direct S wave, the coda, is usually left unmodelled. That coda waves are generated by scattering on crustal heterogeneity is long established [1,2], and finite-difference simulations have shown how random media produce them [3]. Which heterogeneity produces the coda of a particular reservoir, and how strong it must be, remains much less clear, because near-source records with enough bandwidth to test it are rare. Borehole data provide one candidate. Well-log fluctuation spectra in sedimentary, crystalline and volcanic rocks scale approximately as $S(k) \propto k^{-\beta}$ with $\beta \approx 1\text{--}1.5$ over centimetre-to-kilometre scales, and core porosity and permeability are related lognormally [4–6]. A medium

with these statistics (a lognormal, spatially correlated "pink-noise" medium) is the basis of geoflow imaging (GFI) [7,8], and Paper 1 of this series quantified such descriptors at Utah FORGE [9]. Whether the seismic wavefield of FORGE MEQs requires strong small-scale heterogeneity, and whether it can distinguish a power-law spectrum from spectra with a correlation length, has not been tested.

Here we use the 4-kHz records of two downhole sensors in well 56-32 during the April 2022 stimulation and 3-D elastic finite-difference (FD) wavefields for several hundred event–sensor paths per model. We ask three questions, in decreasing order of what the data can answer. (1) Do smooth media, or media with deterministic small-scale structure (thin layers, planar fracture zones), reproduce the late S-wave energy and its inter-event coherence? (2) If not, what effective scattering strength do stochastic media require? (3) Do the waveforms prefer a power-law (pink) spectrum over exponential or Gaussian spectra? We treat the first question as the main test, subject it to controls for window overlap, source, instrument ringing, event selection and numerical convergence, and report the second and third with their limitations. A methodological by-product is a synthetic demonstration that deterministic waveform comparison against an unknown stochastic realisation favours smooth models, so that medium-class selection must use wavefield statistics.

Data

Records and event selection

Table 1 Event selection

Step	Events	Magnitude; distance to sensor mid-point; share of stage 3
GES catalogue, April 2022 stimulation of 16A(78)-32	36 641	—
60-s SEG-2 files read (18–24 April 2022; 4 kHz, 6 channels)	3556 files	—
Largest events with $M \geq -1$ whose 1-s window lies inside a file (extraction cap, ordered by magnitude)	800	−0.44 [−0.61, 0.04]; 317 [262, 414] m; 83 %
Clean P onset on at least one sensor, hypocentre inside the FD interior	555	−0.44 [−0.60, −0.09]; 308 [261, 395] m; 92 %

Clean P onsets on both sensors: per-event coda test, D2, Φ	416	-0.44 [-0.60, -0.11]; 303 [261, 385] m; 97 %
150 highest P-wave SNR (subset of the original analysis; sensitivity only)	150	-0.40 [-0.60, -0.12]; 294 [267, 357] m; 96 %

Median [10th, 90th percentile]. Stage 3 dominates at every step; stage 2 contributes 14 of the 416 events used, and stage 1 none.

Geo-Energie Suisse recorded the April 2022 stimulation of well 16A(78)-32 in 60-s SEG-2 files from monitoring well 56-32 and published a catalogue of 36 641 located events [10]. Fig. 1 shows the geometry, Fig. 2 example records, and Table 1 the selection steps. The only magnitude criterion is the extraction cap of 800 events; no step uses the coda. The per-event test uses all 416 events with clean P onsets on both sensors. The original analysis used the 150 events with the highest P-wave SNR. These have a slightly weaker, not stronger, observed coda than the remaining events (median $\log_{10}$ coda/S -1.33 against -1.12 at sensor A), so the selection did not inflate the coda.

Sensor calibration

The file headers carry no receiver coordinates. Automatic P onsets [11] on the two three-component sensors (channels 1–3, sensor A; 4–6, sensor B) were inverted with the catalogue hypocentres for sensor position (both sensors in one vertical well), clock offsets and V_P (L1 misfit). The solution places the sensors at depths of 2179 m (A) and 2522 m (B) with $V_P = 5808$ m/s. Event-bootstrap 95 % intervals are narrow (± 12 m for A, ± 81 m/s for V_P), but they understate the true uncertainty, because the catalogue locations were themselves computed with a smooth velocity model. Leaving out stimulation stage 3 moves both sensors 52 m deeper and lowers V_P by 8 %, which shows that depth, clock and velocity trade off structurally. We therefore regard sensor depths as uncertain by ~ 50 m and V_P by ~ 8 %, and test the effect of timing errors on the coda measurements directly (see below). The P travel-time residuals of neighbouring events are correlated up to ~ 40 m separation (Fig. 2c), which may reflect structure or correlated catalogue location error; we do not interpret them further.

Channels 1 and 4 carry only $\sim 3\%$ of the P energy and are not used; channels 2–3 and 5–6 behave as horizontal pairs, and all measurements use rotation-invariant horizontal energy.

Sonic log

The 56-32 sonic log (1950–2780 m, calliper-screened) gives a detrended $\ln V$ with $\sigma = 0.045$ at 6-m and 0.032 at 50-m averaging. Its spectral slope depends on band and estimator (Fig. 9). Over 12–200 m the multitaper estimate is $\beta = 0.85 \pm 0.07$ (taper jackknife), and across detrending orders, periodogram, Welch and multitaper estimators and three log segments it ranges from 0.57 to 1.21. Over 1–200 m it is 1.2–1.7. On log-binned spectra a power law is the best of four models over 12–200 m, but exponential and Gaussian autocorrelation spectra with correlation lengths of 2–4 m lie within $\Delta\text{AIC} = 3.4$ and 3.7. Over 1–200 m the Gaussian spectrum is excluded ($\Delta\text{AIC} = 26$), but the exponential one is not ($\Delta\text{AIC} = 2.2$). The log is thus compatible with power-law fluctuations, but a single 830-m vertical log cannot establish scale invariance or exclude a short correlation length. It is also a sample of intact rock along one line, not of the stimulated volume that the MEQ waves traverse.

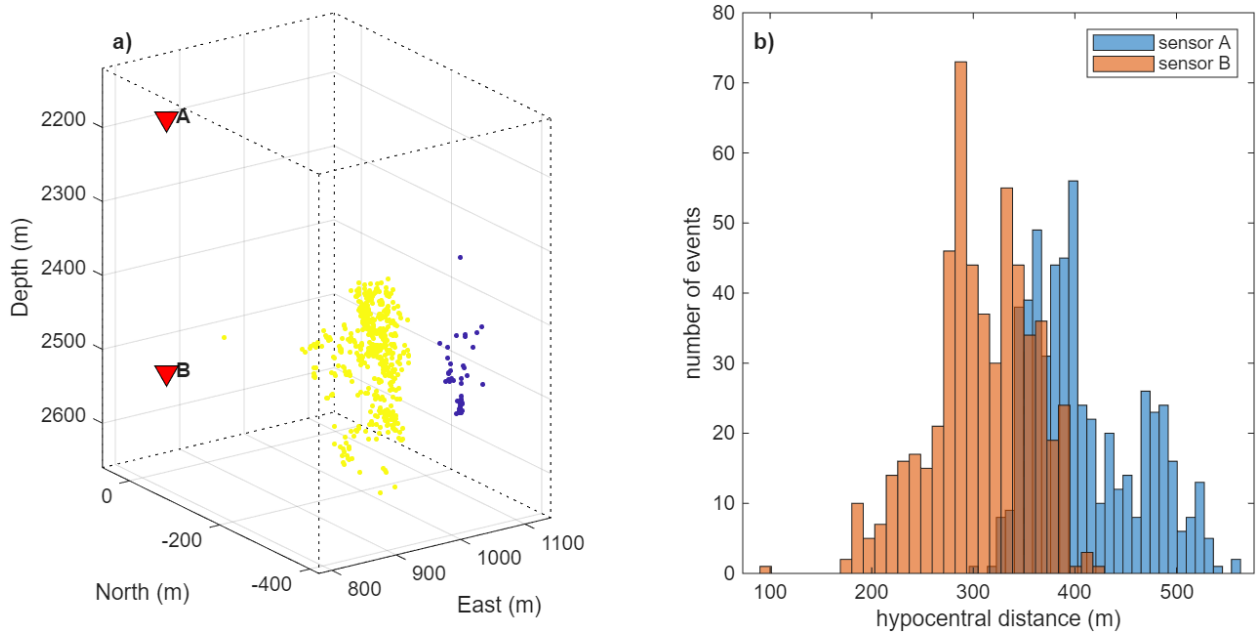


Fig. 1 Utah FORGE 2022 geometry: stimulation MEQs, 56-32 sensors and FD volume

a) Catalogued MEQs inside the FD interior (colour = stimulation stage), the calibrated sensors A and B (red triangles) and the FD interior (dotted box). b) Hypocentral distances to sensors A and B.

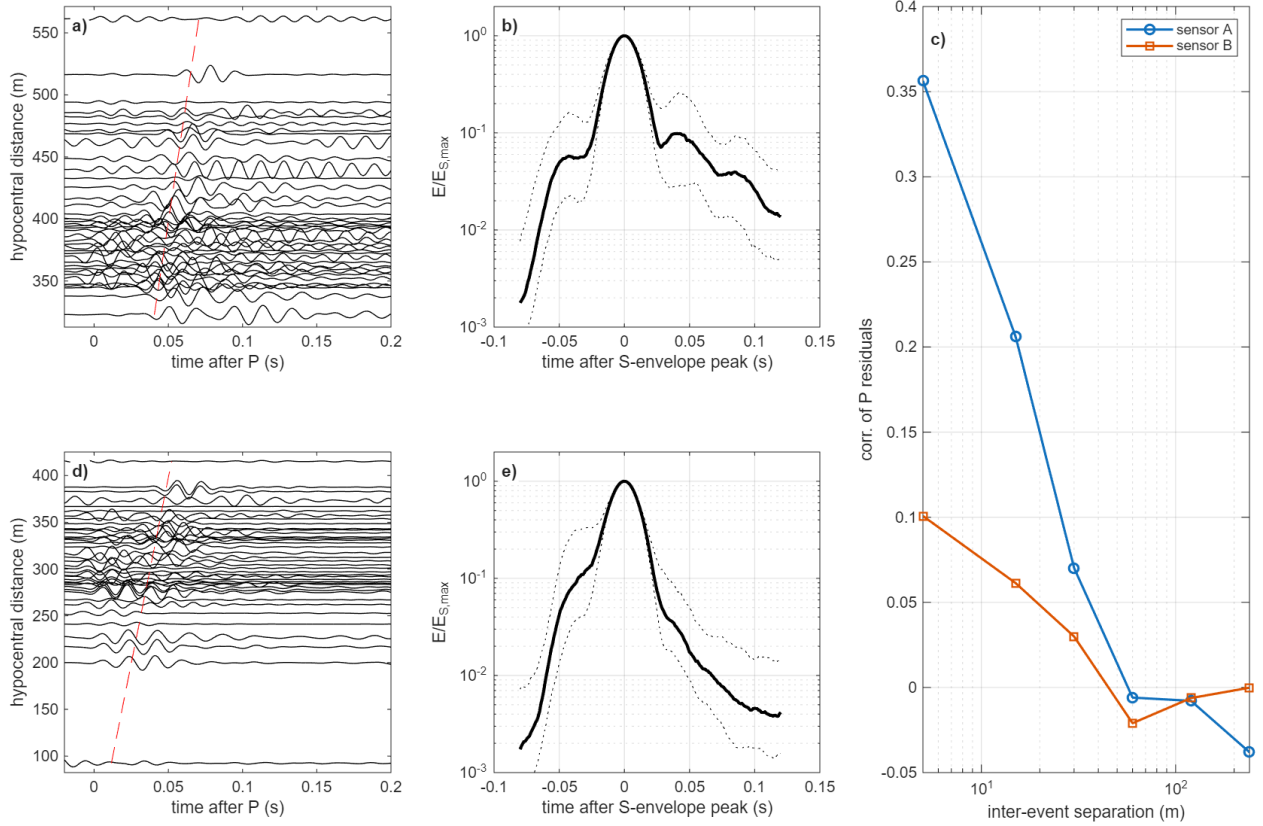


Fig. 2 FORGE 56-32 records and S envelopes

a, d) Horizontal records (one component, 40–80 Hz) of 40 events ordered by hypocentral distance for sensors A and B; the dashed line is the predicted S time. b, e) Median (solid) and interquartile range (dotted) of the horizontal S envelope, 40–80 Hz. c) Correlation of P travel-time residuals versus inter-event separation.

Methods

3-D elastic finite differences and reciprocity

We solve the velocity–stress equations on a staggered grid [12] with fourth-order spatial and second-order temporal differences [13] for a transversely isotropic medium with a vertical symmetry axis (VTI; stiffnesses from V_P , V_S , ρ and Thomsen ϵ , δ , γ [14]). The treatment of

124 anisotropy on the staggered grid follows Igel, Mora and Riollot [15], and the MATLAB
 125 implementation grew out of the acoustic codes of Igel [16], extended to 3-D elastic VTI
 126 media, moment-tensor and point-force sources and strain-rate recording. A Cerjan sponge
 127 [17] absorbs outgoing energy. The grid is $113 \times 137 \times 149$ nodes at 5 m (a $0.36 \times 0.48 \times 0.54$
 128 km interior plus 20-node sponges), with $\Delta t = 0.25$ ms, 1300 steps and a 45-Hz Ricker source.
 129 The analysis band is 20–80 Hz, with ≥ 5.4 grid points per S wavelength at 80 Hz even for the
 130 slowest truncated S velocity of the $\sigma = 0.18$ media (media with $\sigma = 0.22$ and 0.25 , which fall
 131 below 5 points, are shown only in Additional file 1). Three point-force runs per sensor record
 132 the strain-rate tensor at every event node, so that by reciprocity [18] the velocity at sensor
 133 component n from a moment tensor $\mathbf{M}$ at $\mathbf{x}_e$ is

$$v_n(\mathbf{x}_s, t) = M_{pq} \int_0^t \dot{\varepsilon}_{pq}^{(n)}(\mathbf{x}_e, \tau) d\tau. \quad (1)$$

134 Because only horizontal components are used, the two horizontal-force runs per sensor
 135 suffice. Against direct moment-tensor simulations in a heterogeneous VTI medium, the
 136 reciprocity synthetics agree with correlation 0.995 and amplitude ratio 0.995 (Additional file
 137 1: Fig. S1).

138 Coda amplitudes in a small domain could be affected by the grid, the domain edges and the
 139 sponge. We therefore simulated one band-limited stochastic medium ($\sigma = 0.13$, power-law
 140 spectrum with a 780-m outer scale, low-passed at 0.06–0.08 cycles/m so that it is identical on
 141 every grid) four times with sensor B, the same 358 events and the same double-couple
 142 mechanisms: on the reference 5-m grid, on a 2.5-m grid, in a domain extended by 150 m on
 143 every side, and with a sponge of 40 instead of 20 nodes. The coda measures of the three
 144 variants are compared event by event with the reference.

Media

All media have constant density $\rho = 2650 \text{ kg/m}^3$ and background $V_P = 5808 \text{ m/s}$, $V_P/V_S = 1.73$, except where stated. Four smooth media contain no structure below $\sim 50 \text{ m}$: a homogeneous medium (H), a layered medium from the 56-32 sonic log averaged in 50-m blocks (L), a medium with smooth 3-D gradients of $\pm 3 \%$ vertically and $\pm 2 \%$ horizontally across the box (GRAD3), and a homogeneous VTI medium with Thomsen parameters inverted from the P travel times (A). Two deterministic media contain small-scale structure: the sonic log sampled at its 6-m resolution on the 5-m grid (L6), and 12 planar fracture zones, one cell (5 m) thick, about 240 m tall, striking N25°E and dipping 75°, with V_P reduced by 10 % and V_S by 20 % (FZ, which also varies V_P/V_S).

The stochastic media are lognormal, $V(\mathbf{x}) = \bar{V} \exp[\sigma f(\mathbf{x})]$, where f is a zero-mean, unit-variance Gaussian random field obtained by filtering white noise $\hat{w}$ in the wavenumber domain. For the power-law (pink) field the filter is

$$\hat{f}(\mathbf{n}) \propto \frac{\hat{w}(\mathbf{n})}{(1 + |\mathbf{n}|)^p}, |\mathbf{n}| = L|\mathbf{k}|/2\pi, \quad (2)$$

where $\mathbf{n}$ is the dimensionless FFT index, L the box length along each axis and $\mathbf{k}$ the wavenumber. The roll-off wavenumber is therefore $k_0 = 2\pi/L$ (outer scale equal to the box, 0.6–0.75 km), and the power spectrum decays as $|\mathbf{k}|^{-2p}$ above it; $p = 1.5$ gives one-dimensional spectra $\propto k^{-1.05}$ and $p = 1.8$ gives $\propto k^{-1.48}$ along lines through the field. Exponential and Gaussian fields use the filters $(1 + k^2 a^2)^{-1}$ and $\exp(-k^2 a^2/8)$ with the correlation length a in metres. All fields are periodic over the FD box, which is larger than any source–receiver path. After scaling, $\ln V$ is truncated at $\pm 3\sigma$ ($\pm 2.5\sigma$ for $\sigma > 0.15$) without renormalisation. Truncation affects 0.25–1.2 % of cells and reduces σ by $\leq 1.1 \%$ (Additional file 1: Table S9). The truncated velocity range is wide: at $\sigma = 0.13$, V_P spans 3.9–8.6 km/s, and at $\sigma = 0.18$ 3.7–9.1 km/s. The extremes occupy a small fraction of the volume, but they show that $\sigma \geq 0.13$ is not a small perturbation.

In the default media V_P and V_S share one field, so V_P/V_S is constant. Two variants relax this at $\sigma = 0.13$: independent fields for V_P and V_S (VSD), and a V_S -dominated medium with $\sigma(\ln V_S) = 0.13$ and $\sigma(\ln V_P) = 0.065$ (VSO). Because σ absorbs any unmodelled contribution to the late energy, including density contrasts, anisotropic fracture scattering, intrinsic attenuation and site effects, we call the fitted value an *effective scattering-strength parameter* σ_{eff} within this isotropic model class, not a measured property of the rock.

Media were simulated in two sets (Table 2). The original set explores σ and p for the pink family. The revision adds a common grid for all three families: $\sigma = 0.09, 0.13$ and 0.18 ; $p = 1.5$ and 1.8 (pink), $a = 15$ and 50 m (exponential, Gaussian); two white-noise seeds each (11 and 33). Every family thus has the same number of candidates on the common grid. Fields of different families generated from the same seed share their phases and are strongly correlated (the pink and $a = 15$ m exponential fields with seed 33 by 96 %); this is a controlled comparison of spectral shape, not a test with independent fields.

Table 2 Media simulated

Model	Class	Parameters
H	smooth: homogeneous	V_P 5808 m/s, V_P/V_S 1.73
L	smooth: 1-D layered	56-32 sonic log, 50-m blocks
GRAD3	smooth: 3-D gradients	± 3 % vertical, ± 2 % E and N
A	smooth: homogeneous VTI	travel-time inverted ϵ , δ
L6	deterministic: 1-D layered	56-32 sonic log at 6-m resolution
FZ	deterministic: fracture zones	12 planar zones, 5 m thick, N25°E/75°, V_P -10 %, V_S -20 %
P4, P1, P3, P6, P9	pink, $p = 1.5$ (seed 11)	$\sigma = 0.02, 0.045, 0.09, 0.13, 0.18$
T1, R1, R2; R3; R6; R5	pink, $p = 1.5$	further seeds at $\sigma = 0.045; 0.09; 0.13; 0.18$
P5, P2; P7, P8; P11	pink, other p	$p = 1.2, 1.8$ at $\sigma = 0.045, 0.09$; $p = 1.8$ at 0.18
P12, P10	pink, $p = 1.5$	$\sigma = 0.22, 0.25$ (Additional file 1 only)
Common grid	pink, exponential, Gaussian	$\sigma = 0.09, 0.13, 0.18 \times (p = 1.5, 1.8 \text{ or } a = 15, 50 \text{ m}) \times \text{seeds } 11, 33$
VSD, VSO	pink, variable V_P/V_S	$\sigma = 0.13$; independent V_S field; $\sigma_{VS} = 0.13$, $\sigma_{VP} = 0.065$
R6c, R6d, E15c/d, G50c/d	synthetic truths	$\sigma = 0.13$, seeds 44 and 55 (shape-resolution test)
T2	synthetic truth	50-m layering + VTI ($\epsilon = 0.06$, $\delta = 0.03$, $\gamma = 0.06$)

All media: $\rho = 2650 \text{ kg/m}^3$. Stochastic media: lognormal, truncated at $\pm 3\sigma$ ($\pm 2.5\sigma$ for $\sigma > 0.15$). The common grid comprises 36 media (12 per family), 14 of which already existed in the original set.

Source and preprocessing

FD synthetics carry a 45-Hz Ricker source. To give data and synthetics the same source spectrum, each observed trace is convolved with the same Ricker wavelet. This is valid if the true source spectrum is flat in the analysis band, i.e. if corner frequencies exceed ~ 80 Hz. The S-window displacement spectra (raw data, 5–300 Hz) support this for most but not all events. The apparent corner, the first frequency above 50 Hz at which the spectrum falls 6 dB below its 30–50 Hz level, has a median of 128 Hz at sensor A and 90 Hz at B. Because it includes path attenuation, it is a lower bound on the source corner; 21 % (A) and 34 % (B) of events have apparent corners below 80 Hz. To bound the effect of finite source duration, we convolved the homogeneous and $\sigma = 0.13$ pink synthetics with Brune source-time functions with corners of 150, 100 and 60 Hz. Even the 60-Hz source raises the homogeneous coda-to-S ratio by only 0.35 log units (A) and changes the pink one by +0.03. Source duration therefore cannot close a deficit of 1–2 log units.

Per-event coda prediction test

For each event and medium, the moment tensor is fitted by linear least squares to the four horizontal traces (two sensors, 40–80 Hz) in the window $P - 10$ ms to $S + 15$ ms, with a per-sensor time shift of up to ± 6 ms and the tool azimuth and handedness of each sensor fixed from a previous grid search. The fitted moment tensor then predicts the whole record. After a guard band of 25 ms, the test quantity is the ratio of mean horizontal energy in $S + 40$ to $S + 100$ ms to the peak S energy ($S - 8$ to $S + 15$ ms), compared per event between prediction and observation as $\log_{10}(\text{predicted/observed})$. The fit and coda windows do not overlap. In the original version of this analysis the fit window extended to $S + 60$ ms and overlapped 43 % of the coda window; that version is superseded. The sensitivity analysis varies the end of the fit window ($S + 10$, 15, 20 and, for comparison, the original 60 ms) and the coda window ($S + 30$ –100, 40–100 and 50–110 ms). It also compares a full moment tensor, a deviatoric one and

210 a double couple (grid search over strike, dip and rake, with a least-squares scalar moment).
211 Condition numbers of the column-normalised design matrix and the ISO, CLVD and DC
212 fractions of the full solutions are reported. A timing check compares the observed S-envelope
213 peak with the predicted S time. The test is repeated on the events whose S peak lies within
214 ± 10 ms (and ± 5 ms) of the prediction on both sensors.

215 **Tests for instrument and borehole ringing**

216 A resonant sensor or borehole response $h(t)$ moves energy from the S peak into later
217 windows, so a coda-to-S ratio does not cancel it. We applied three tests. First, the S-window
218 spectra and the coda/S spectral ratio were examined for resonance peaks. In the analysis band
219 (30–120 Hz), any peak in the smoothed S spectrum is at most 0.35 dB above its surroundings.
220 The same measure detects the resonances near 330, 480 and 920 Hz (prominence up to 10
221 dB), which serve as a positive control. Second, in a null-ringing test the homogeneous
222 synthetics were convolved with the strongest single resonance (40–80 Hz, $Q = 5$ –40) whose
223 spectral peak does not exceed the observed in-band prominence, and the coda-to-S ratio was
224 recomputed. Third, the two sensors, which have different coupling and different in-band
225 peaks, are analysed separately throughout. No calibration shots, perforation shots or third
226 sensor were available for 2022, so an empirical transfer function could not be measured.

227 **Statistical comparison of media**

228 With two sensors the realisation of a random medium cannot be recovered, so stochastic
229 media are compared with the data through wavefield statistics. On the horizontal energy
230 envelope we measure three S-normalised features per event: the coda level $\log_{10}(E_{\text{coda}}/E_S)$ 30–
231 100 ms after the S peak, the coda decay rate and the rms S-pulse width. We use three sensor–
232 band combinations: A 20–40 Hz, A 40–80 Hz and B 40–80 Hz. B 20–40 Hz is excluded
233 because it carries a persistent non-event plateau. The misfit is

$$\begin{aligned}
\Phi = & \frac{1}{9} \sum_{c=1}^3 \sum_{j=1}^3 D_{\text{KS}}(F_{cj}^{\text{obs}}, F_{cj}^{\text{mod}}) \\
& + \frac{1}{3} \sum_{c=1}^3 \left[\frac{1}{N_\tau} \sum_{\tau=0}^{100\text{ms}} \left(\log_{10} \tilde{E}_c^{\text{obs}}(\tau) \right. \right. \\
& \left. \left. - \log_{10} \tilde{E}_c^{\text{mod}}(\tau) \right)^2 \right]^{1/2}
\end{aligned} \tag{3}$$

234 where D_{KS} is the two-sample Kolmogorov–Smirnov distance between the observed and
 235 synthetic distributions of feature j for combination c , and $\tilde{E}_c$ is the median post-peak S
 236 envelope. All nine KS terms and all three envelope terms are weighted equally, events are
 237 weighted equally, and events lacking a measurement in one combination are omitted from that
 238 combination only. The features are correlated, so Φ is a ranking statistic, not a likelihood. Its
 239 uncertainty is estimated by an event bootstrap (200 resamples, the same resample for every
 240 medium). Family comparisons use the best member of each family on the common grid per
 241 bootstrap sample, and a holdout test in which each family's best (σ , shape, seed) is chosen on
 242 half of the events and scored on the other half (20 random splits plus a near/far split by
 243 distance).

244 **Inter-event coda coherence**

245 For pairs of events recorded on the same sensor, D2 is the maximum normalised two-
 246 component cross-correlation (lag ± 8 ms) of the 40–80 Hz window $S + 15$ to $S + 100$ ms,
 247 summarised as the median in bins of inter-event separation (0–5 to 80–160 m). Pairs that
 248 share an event are not independent, so 95 % intervals come from a cluster bootstrap over
 249 events (each resampled event carries all its pairs). To test sensitivity to catalogue location
 250 error, the separations were recomputed after perturbing every hypocentre by 5 and 10 m.

Synthetic truth tests

The ability of the statistical comparison to select a medium class was tested with synthetic truths. Two truths were used in the main test: a pink medium (T1, $\sigma = 0.045$) and 50-m layering with VTI (T2). A further nine truths at equal $\sigma = 0.13$ (pink, exponential $a = 15$ m and Gaussian $a = 50$ m, seeds 33, 44 and 55) tested resolution of the spectral family. Synthetic records use double-couple mechanisms consistent with the FORGE stress field, and real noise at each event's observed SNR.

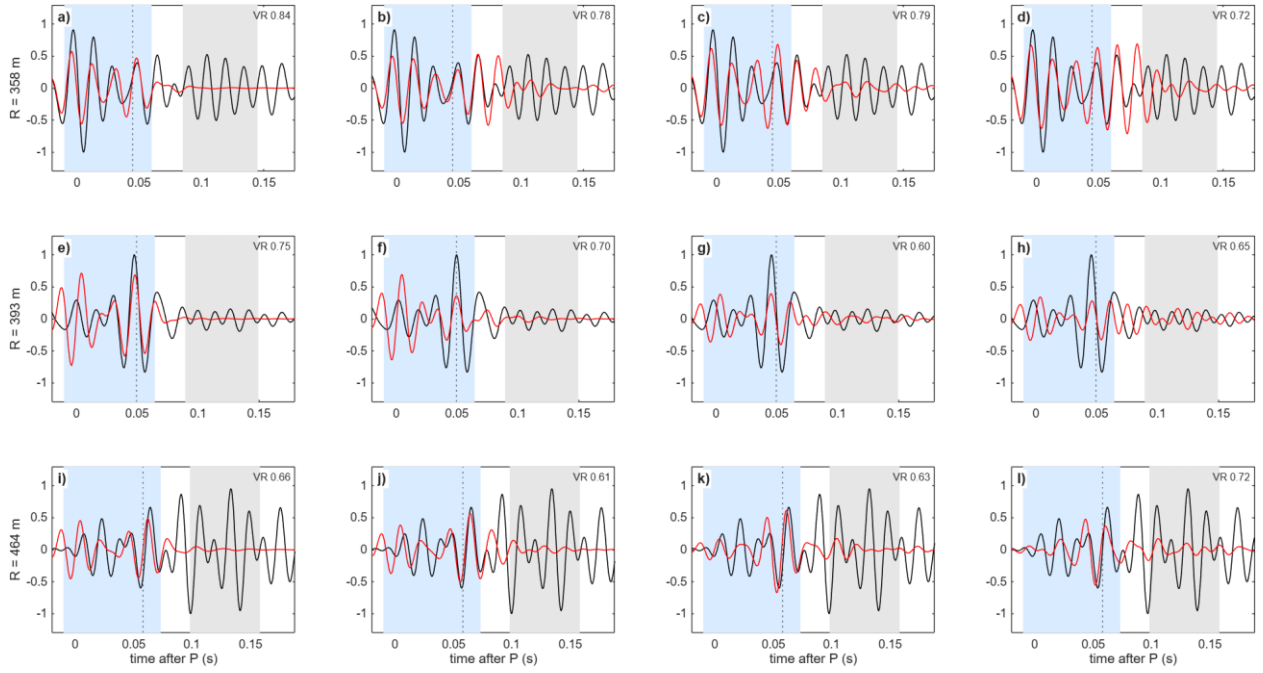
Results

Smooth media under-predict the late S energy

Fig. 3 compares observed and predicted sensor-A seismograms for three events. The moment tensor fitted to the direct waves reproduces the P and S pulses in all media. In the homogeneous medium the prediction then decays to near zero within the guard band, whereas the observed traces keep oscillating through the coda window. Table 3 and Fig. 4 give the statistics over 416 events. The four smooth media under-predict the coda-to-S ratio by a median factor of 44–82 at sensor A and 12–15 at sensor B, and only 1–5 % (A) and 5–11 % (B) of events fall within a factor of three. Smooth 3-D gradients ($-1.67/-1.11$, A/B) and VTI anisotropy ($-1.91/-1.19$) alter the direct waves but add no late energy.

Deterministic small-scale structure changes this substantially. The sonic log at its 6-m resolution (L6) gives $-1.28/-0.53$, and the thin fracture zones (FZ) give $-1.04/-0.39$: under-prediction by factors of 19 and 11 at sensor A but only 3.4 and 2.4 at sensor B, where 41 % and 45 % of events fall within a factor of three. Averaging the log in 50-m blocks, as in the original analysis, therefore removed most of the late energy that the logged layering can generate. For the level of late energy, structure at the 5–10 m scale is the essential ingredient, whether it is layered, planar or random. Stochastic media span the observations: pink media

275 with $\sigma = 0.02$ under-predict ($-1.20/-0.49$), $\sigma = 0.09$ is close at sensor A (-0.14) and $\sigma = 0.045$
276 at sensor B ($+0.02$), and the seed-11 stochastic media with $\sigma = 0.13$ over-predict ($+0.28$ to
277 $+1.12$). The best stochastic media bring 53 % (A) and 54 % (B) of events within a factor of
278 three. The residual scatter reflects the unknown realisation. No single medium matches the
279 median at both sensors; for the pink family, sensor B requires about half the σ of sensor A.



280
281 **Fig. 3 Observed and predicted sensor-A seismograms with non-overlapping fit and coda**
282 **windows**

283 Horizontal component 1, 40–80 Hz, for three events at increasing distance (rows; distance on
284 the left). Columns: homogeneous medium, deterministic fracture zones (FZ), pink media with
285 $\sigma = 0.045$ and 0.09 . Black: observed; red: predicted with the moment tensor fitted in the light-
286 blue window ($P - 10$ ms to $S + 15$ ms); grey: coda window ($S + 40$ to $S + 100$ ms); dotted:
287 predicted S time. Variance reduction of the fit in each panel.

288 **Table 3 Predicted versus observed late S energy for 416 FORGE events**

Medium	Median $\log_{10}(\text{pred}/\text{obs})$, A	Median, B	% within $\times 3$, A	% within $\times 3$, B	VR
smooth: homogeneous	-1.80	-1.09	2	9	0.75
smooth: layered, 50 m	-1.64	-1.17	5	5	0.73
smooth: 3-D gradients	-1.67	-1.11	3	9	0.74
smooth: VTI	-1.91	-1.19	1	11	0.64

deterministic: layered, 6 m	-1.28	-0.53	15	41	0.73
deterministic: fracture zones	-1.04	-0.39	24	45	0.73
pink $\sigma = 0.02$	-1.20	-0.49	16	42	0.73
pink $\sigma = 0.045$	-0.74	+0.02	34	51	0.72
pink $\sigma = 0.09$	-0.14	+0.53	50	35	0.67
pink $\sigma = 0.13$	+0.31	+0.78	50	26	0.62
pink $\sigma = 0.18$	+0.61	+1.03	37	18	0.60
pink $\sigma = 0.13$, independent V_s	-0.13	+0.54	53	35	0.66
pink, V_s -dominated	+0.32	+0.79	49	24	0.62
exponential $a = 15$ m, $\sigma = 0.09$	-0.07	+0.61	51	33	0.68
exponential $a = 15$ m, $\sigma = 0.13$	+0.28	+0.94	50	22	0.64
exponential $a = 50$ m, $\sigma = 0.09$	-0.16	+0.39	48	41	0.65
exponential $a = 50$ m, $\sigma = 0.13$	+0.31	+0.71	45	27	0.59
Gaussian $a = 15$ m, $\sigma = 0.09$	+0.14	+0.79	48	22	0.69
Gaussian $a = 15$ m, $\sigma = 0.13$	+0.42	+1.12	43	15	0.64
Gaussian $a = 50$ m, $\sigma = 0.09$	-0.45	-0.35	39	44	0.62
Gaussian $a = 50$ m, $\sigma = 0.13$	+0.29	+0.54	45	33	0.56

Late energy = mean horizontal energy $S + 40$ to $S + 100$ ms divided by the S-peak energy, 40–80 Hz. Moment tensors fitted in $P - 10$ ms to $S + 15$ ms (no overlap with the coda window). VR = median variance reduction of the direct-wave fit. Stochastic media: seed 11; other seeds in Additional file 1.

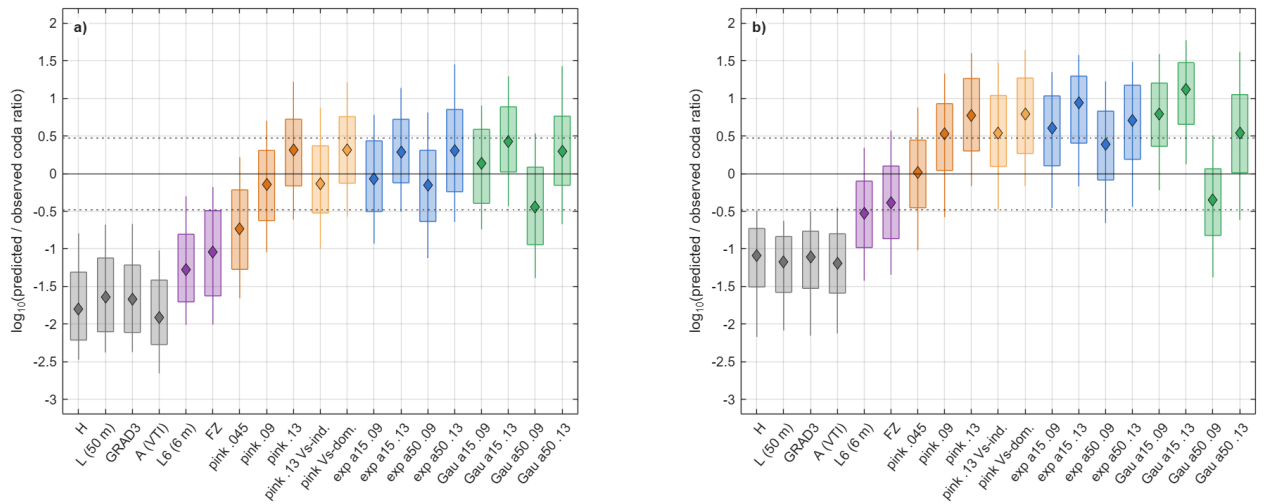


Fig. 4 Per-event predicted/observed late S energy for all media

a) Sensor A. b) Sensor B. 416 events, 40–80 Hz, windows as in Table 3. Boxes: interquartile range; whiskers: 10–90 %; diamonds: median; dotted lines: factor of three. Grey: smooth media; purple: deterministic small-scale media (6-m layering, fracture zones); orange: pink media (light orange: variable V_p/V_s); blue: exponential; green: Gaussian.

Robustness of the deficit

Table 4 Robustness of the late-energy deficit of smooth and deterministic media

Variant	Smooth, A	Smooth, B	L6 and FZ,	L6 and FZ,
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	A		B	
Primary (fit to S + 15 ms; coda S + 40–100 ms)	–1.91 to –1.64	–1.19 to –1.09	–1.28 to –1.04	–0.53 to –0.39
Fit to S + 10 ms	–1.91 to –1.64	–1.20 to –1.07	–1.29 to –1.04	–0.52 to –0.33
Fit to S + 20 ms	–1.91 to –1.64	–1.19 to –1.08	–1.28 to –1.07	–0.53 to –0.39
Coda S + 50–110 ms	–2.21 to –1.99	–1.44 to –1.39	–1.46 to –1.33	–0.65 to –0.64
Coda S + 30–100 ms	–1.58 to –1.45	–1.05 to –0.99	–1.11 to –0.82	–0.48 to –0.19
Original overlapping fit (S + 60 ms), coda S + 30–100 ms	–1.59 to –1.45	–1.03 to –0.99	–1.12 to –0.93	–0.50 to –0.29
Deviatoric moment tensor	–2.04 to –1.69	–1.21 to –1.09	–1.33 to –1.19	–0.51 to –0.49
Double couple	–2.10 to –1.69	–1.25 to –1.11	–1.34 to –1.24	–0.58 to –0.51
150 highest-SNR events	–1.86 to –1.53	–1.26 to –1.09	–1.22 to –0.77	–0.65 to –0.33
S peak within ± 10 ms on both sensors (250 events)	–1.70 to –1.42	–1.17 to –0.98	–1.10 to –0.88	–0.53 to –0.35
S peak within ± 5 ms on both sensors (168 events)	–1.62 to –1.35	–1.20 to –0.98	–1.04 to –0.87	–0.55 to –0.42
Homogeneous + worst in-band resonance (null-ringing test)	–1.99 to –1.61	–1.33 to –0.96		
Homogeneous, Brune source $f_c = 60$ Hz (change)	+0.35	+0.35		

300 Median $\log_{10}(\text{predicted/observed late energy})$; range over the smooth media (H, L, GRAD3, A) and over the
301 deterministic small-scale media (L6, FZ). Null-ringing test: range over resonance frequencies 40–80 Hz and $Q =$
302 5–40, for the median over events of the homogeneous double-couple synthetics relative to the observed median.
303 Brune row: change of the homogeneous coda-to-S ratio relative to an impulsive source.
304 Table 4 shows that the deficit does not depend on the analysis choices. Ending the fit window
305 at S + 10 or S + 20 ms, or moving the coda window, changes the smooth-model medians by a
306 few tenths of a log unit. For the smooth media the window overlap of the original analysis had
307 little effect: with the same coda window, fitting to S + 60 instead of S + 15 ms changes the
308 medians by at most 0.02. The original coda window, which began at S + 30 ms, gave a
309 smaller deficit than the corrected one simply because it included more of the early coda.
310 Deviatoric and double-couple solutions give the same result, although they fit the direct
311 waves less well (median VR in the homogeneous medium 0.75 for the full, 0.71 for the
312 deviatoric and 0.67 for the double-couple solution). The moment-tensor problem is
313 moderately conditioned (median condition number 16–18 for smooth and 4–16 for stochastic

media, decreasing with σ). The full solutions carry large non-double-couple parts (median $|\text{ISO}|$ 28 %, $|\text{CLVD}|$ 28 % in the homogeneous medium). With two sensors in one well these parts are poorly resolved and we do not interpret them. The point is that no source constraint changes the coda deficit. The observed S peak lies within ± 10 ms of the predicted time for 250 events on both sensors, and the deficit remains in that subset and in the ± 5 ms subset. Timing and location errors therefore do not explain it.

The ringing tests constrain the instrument explanation (Fig. 5). Between 30 and 120 Hz the S spectra of both sensors are smooth, and the coda/S spectral ratio has no peak. The worst resonance compatible with this smoothness leaves the homogeneous medium 1.6–2.0 log units short at sensor A and 1.0–1.3 at B. The resonances above 300 Hz are strong and clearly visible, so an in-band resonance large enough to create the observed coda would not have escaped detection. This test excludes a single resonant mode, but not a broadband, non-resonant coupling response with a long impulse response. That possibility can be excluded only with an empirical transfer function, which is not available.

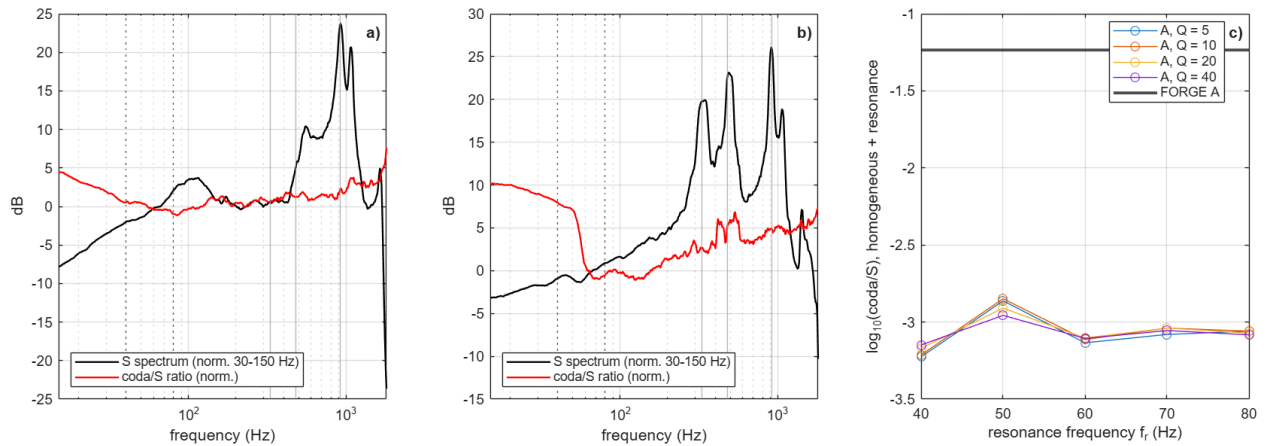


Fig. 5 Tests for instrument or borehole ringing

a, b) Sensors A and B: median S-window spectrum normalised over 30–150 Hz (black) and median coda/S spectral ratio (red, normalised). Dotted lines mark the analysis band 40–80 Hz and grey lines the resonances near 330, 480 and 920 Hz. c) Null-ringing test: $\log_{10}$ coda/S of the homogeneous synthetics (sensor A) after convolution with the strongest resonance

compatible with the observed in-band spectral smoothness, versus resonance frequency, for Q
= 5–40. Thick line: observed FORGE median.

Grid refinement and sponge thickness change the late energy little (Additional file 1: Table
S10). Relative to the 5-m reference, the median coda-to-S ratio at 40–80 Hz changes by +0.01
log units on the 2.5-m grid and by +0.13 with the doubled sponge (+0.16 and –0.13 at 20–40
Hz). Enlarging the domain by 150 m on every side, however, raises it by +0.40 at 40–80 Hz
(+0.09 at 20–40 Hz) and lowers the coda decay rate from 48 to 21 s^{–1}. Because the doubled
sponge has a much smaller effect, this is not a boundary artefact. It reflects the larger
scattering volume: in the reference domain the event cloud nearly fills the interior, so
heterogeneity beyond it lies in the absorbing sponge and returns no scattered energy. A
smooth medium has no such scatterers, so the smooth-model deficit is unaffected. The slower
decay in the larger domain may also explain why the observed coda decayed more slowly
than the simulated coda in the original analysis. For the stochastic media, the reference
domain underestimates the late energy, and the effective scattering strength derived below is
an upper bound. With the sensitivity of the median ratio to σ found below (0.80 log units per
unit change of $\ln \sigma$), the domain effect corresponds to σ_{eff} values about 1.7 times too large.

Effective scattering strength

For each family the σ at which the median predicted late energy equals the observed one (σ_{eff})
is interpolated in $\ln \sigma$ between the simulated levels of the reference domain (Fig. 6;
Additional file 1: Table S11). At sensors A and B it is 0.104 and 0.053 for pink media with p
= 1.5, 0.098 and 0.048 with $p = 1.8$, 0.093 and < 0.09 for exponential media with $a = 15$ m,
0.102 and < 0.09 with $a = 50$ m, and 0.112 and 0.131 for Gaussian media with $a = 50$ m ("
0.09": the smallest simulated σ already over-predicts). σ_{eff} thus differs between sensors and
families, so it is a model-dependent summary of the late energy rather than a unique rock
property. Media with less small-scale content need larger σ (Gaussian $a = 50$ m), and media

richest in structure near the S wavelength need less (Gaussian $a = 15$ m: < 0.09 at both sensors). Realisation scatter adds to this. The two seeds at one σ differ in the median ratio by up to 0.47 log units for pink media and 0.40 for exponential media with $a = 15$ m, i.e. a factor of about 1.8 in σ_{eff} , but by up to 1.57 for Gaussian media with $a = 50$ m, whose few large features make each realisation idiosyncratic in a 0.5-km box.

Relaxing the constant V_P/V_S ratio changes the late energy only moderately. With independent V_P and V_S fields the median ratio is $-0.13/+0.54$ (A/B), against $+0.31/+0.78$ for the coupled medium. The V_S -dominated medium, with half the V_P perturbation, gives $+0.32/+0.79$. The S coda is controlled mainly by the V_S perturbation, so σ_{eff} should be read as a V_S scattering strength.

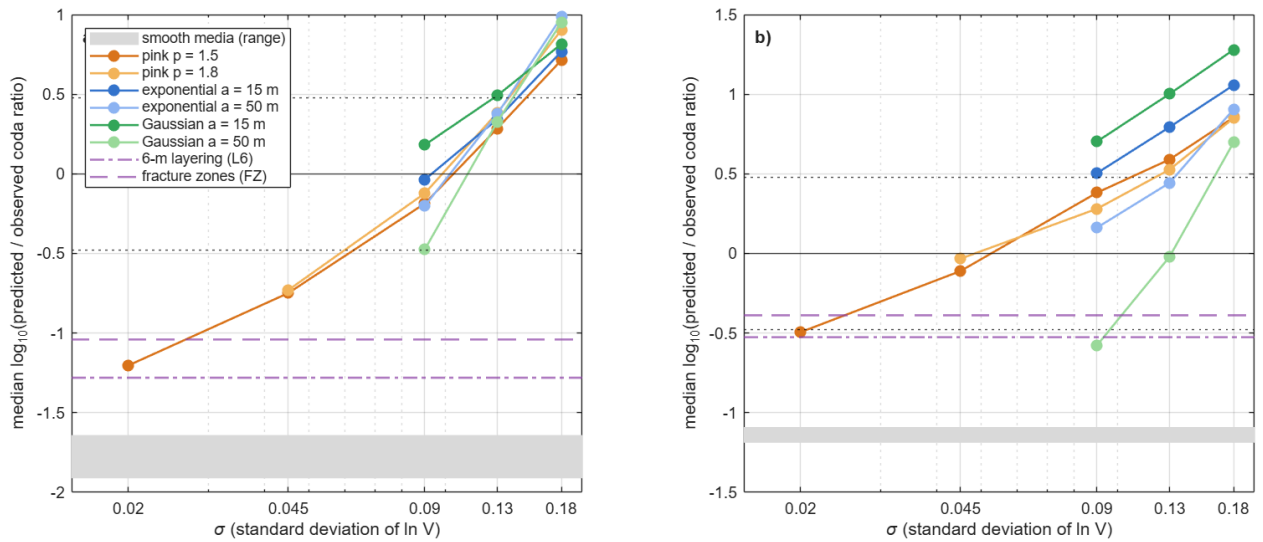


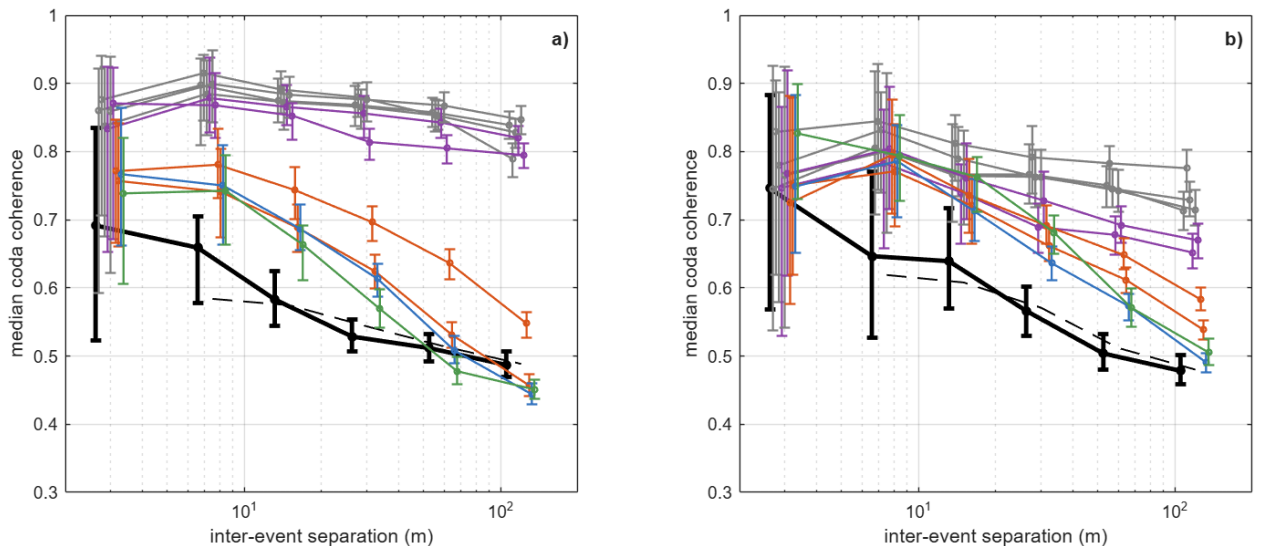
Fig. 6 Median predicted/observed late S energy versus σ for each spectral family

a) Sensor A. b) Sensor B. Median over events of $\log_{10}(\text{predicted/observed})$, averaged over realisations where several exist. The zero crossing defines σ_{eff} ; dotted lines: factor of three; grey band: range of the smooth baselines.

Inter-event coda coherence

Fig. 7 shows D2 with cluster-bootstrap intervals. The observed coda of neighbouring events is moderately coherent (0.69 at sensor A and 0.75 at B for 0–5 m) and decorrelates to 0.49 and

377 0.48 at 80–160 m. In the smooth media the coda remains coherent (0.79–0.85 at 80–160 m,
 378 sensor A; 0.71–0.78 at B), and their 95 % intervals lie above the data interval in every bin
 379 from 5 m separation at sensor A and from 20 m at sensor B. The deterministic small-scale
 380 media stay as coherent as the smooth media at sensor A (0.79–0.82) and decorrelate only
 381 partly at sensor B (0.65–0.67); their intervals lie above the data from 5 m (A) and 20 m (B).
 382 Thin layers and planar fracture zones thus add late energy, but energy that is similar for
 383 neighbouring events, because it is generated by the same few structures along similar paths.
 384 Stochastic media decorrelate as the data do, although most stay more coherent than the data at
 385 10–40 m. Perturbing the hypocentres by 10 m changes the observed medians by ≤ 0.08 . The
 386 0–5 m bin contains only 64 pairs from 96 events, and its interval is wide. D2 is measured on
 387 the same records as the late energy, so it is a second descriptor of the same data, not an
 388 independent experiment. It does, however, discriminate between the media that the late-
 389 energy level alone cannot separate: it requires the scattering structure to vary on the scale of
 390 the event separations, as 3-D random heterogeneity does and a few deterministic layers or
 391 planes do not.



392

393 **Fig. 7 Inter-event coda coherence with cluster-bootstrap uncertainty**

394 a) Sensor A. b) Sensor B. Median coherence of the 40–80 Hz coda ($S + 15$ to $S + 100$ ms)
 395 versus inter-event separation, with 95 % cluster-bootstrap intervals over events. Black:

396 FORGE; dashed: FORGE with hypocentres perturbed by 10 m; grey: smooth media (H, L,
397 GRAD3, A); purple: 6-m layering (L6) and fracture zones (FZ); orange: pink ($\sigma = 0.09, 0.13$);
398 blue: exponential ($a = 15$ m); green: Gaussian ($a = 50$ m), $\sigma = 0.13$. Symbols are offset
399 horizontally for clarity.

400 **Statistical comparison and the spectral family**

401 In the synthetic tests the statistical comparison selected the generating class in both controlled
402 examples. With the pink truth T1 the best medium was a pink medium with the true σ ($R1, \Phi$
403 $= 0.09$), ahead of H, L and A ($\Phi = 0.29$ each). With the layered-VTI truth T2, L and A ranked
404 first ($0.13, 0.16$) and every pink medium was rejected. Two examples do not establish that the
405 procedure is unbiased. The deterministic moment-tensor fit behaved differently. For the pink
406 truth it preferred the smoothest media (median VR 0.93 for H against 0.88 for a pink medium
407 with the correct statistics but another realisation), because a wrong realisation places scattered
408 energy at wrong times. This applies to selecting a model class from a set of arbitrary
409 stochastic realisations. It does not imply that full-waveform inversion, which updates the
410 structure itself, cannot recover heterogeneity when the data coverage allows it.

411 The equal- σ family test selected the correct family for 5 of 9 truths. Under a chance model
412 with $p = 1/3$ the probability of at least 5 successes is 0.145, so this is not significant. Pink and
413 exponential truths were assigned to the correct member of that pair in 3 of 6 cases (chance
414 level), whereas small-scale-rich (pink or exponential) and Gaussian truths were separated in 8
415 of 9. With so few realisations, even the latter is only indicative.

416 On the FORGE data (Fig. 8), the smooth media (best $\Phi = 1.33$) and the deterministic small-
417 scale media (1.01) are separated from every stochastic family (best members 0.25–0.57) by
418 far more than the bootstrap intervals. Between the stochastic families, the best single member
419 belongs to the exponential $a = 50$ m family ($\Phi = 0.25$ [0.24, 0.30]), which has the lowest Φ in
420 98 % of event-bootstrap resamples, ahead of the pink family (0.29 [0.27, 0.34]). In the

Sonic-log spectrum

Fig. 9 shows the log spectrum with four fitted spectral models. The spectrum is consistent with a power law over 12–200 m, but this range spans only 1.2 decades. Short-correlation-length spectra fit almost as well, and the estimated slope depends on estimator and segment. The log therefore does not select a spectral family independently either.

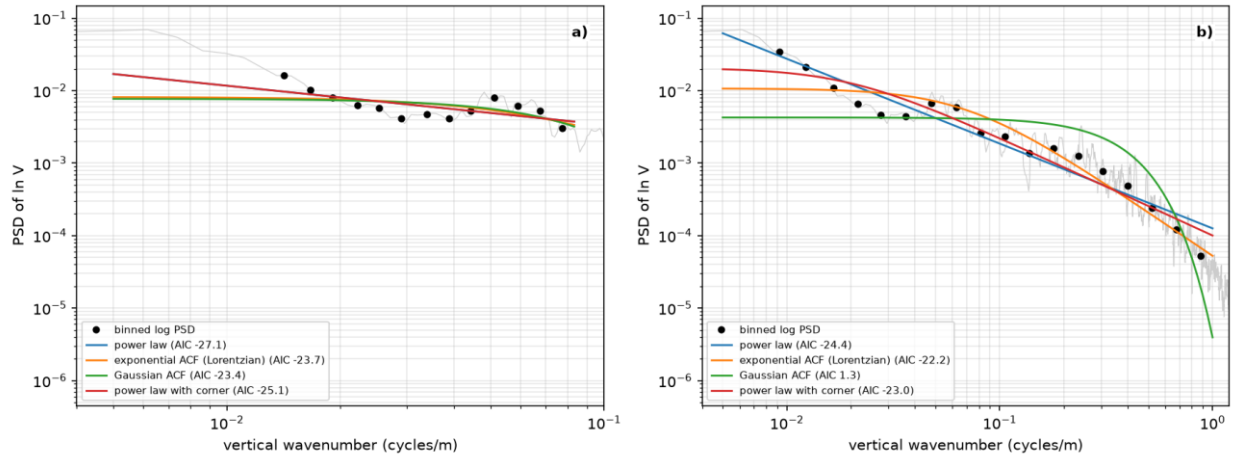


Fig. 9 Spectral models of the 56-32 sonic-log fluctuations

Multitaper power spectrum of detrended $\ln V$ (grey) and log-binned values (dots), with least-squares fits of a power law, exponential and Gaussian autocorrelation spectra and a power law with a corner. a) 12–200 m; the corner model converges to a pure power law and overlies it. b) 1–200 m. AIC values in the legend.

Discussion

What the data show

Media without structure below ~ 50 m (homogeneous, 50-m layering, smooth 3-D gradients, VTI) produce one to two orders of magnitude less late S energy than recorded, event by event. This deficit survives non-overlapping windows, alternative source constraints, finite source duration, S-timing selection and a bounded single-resonance ringing model, and the simulated coda is converged with respect to grid and absorbing boundary. Deterministic structure at the

5–10 m scale, either the logged layering at full resolution or thin planar fracture zones, produces much of the missing energy, particularly at the deeper sensor. Such structure keeps the coda of neighbouring events coherent, however, whereas the observed coda decorrelates. Stochastic 3-D media reproduce both the level and the decorrelation. What remains true if the term "pink noise" is removed from this paper is therefore: within elastic modelling, the late S energy and its decorrelation in the 56-32 records require strong small-scale 3-D heterogeneity, unless an unmodelled site or coupling response of comparable size is present.

What the data do not show

First, the waveforms do not identify the spectral family. At 20–80 Hz, power-law, exponential and Gaussian media reach comparable agreement once σ is adjusted, and the equal- σ truth tests do not separate pink from exponential media. The original analysis had suggested that the 100-Hz coda level (2.5-m grid, sensor B) rejects a large-scale Gaussian medium. That comparison used fixed $\sigma = 0.13$, two Gaussian realisations and noise-free synthetics, and changed grid, domain and sensor at once. We therefore report it only as an exploratory result (Additional file 1: Fig. S9). The same applies to the coda-attenuation diagnostics (Q_c^{-1}): at 25 Hz the coda window spans one to three periods and even the homogeneous medium yields a non-zero Q_c^{-1} . We treat them as finite-window diagnostics and draw no inference from the number of octave bands (Additional file 1: Figs. S7, S8).

Second, σ_{eff} is not a unique measure of rock heterogeneity. It absorbs every unmodelled contribution to the late energy: density contrasts, anisotropic or fluid-filled fracture scattering, intrinsic attenuation (which would reduce the coda and raise σ_{eff}), and site or coupling effects that the ringing test cannot exclude. It also depends on the simulation domain. In the reference domain $\sigma_{\text{eff}} = 0.05$ – 0.13 , depending on sensor and family. The domain test suggests that these values are about 1.7 times too large. In the original analysis, the misfit optimum ($\sigma \approx 0.18$) and the per-event optimum ($\sigma \approx 0.09$ – 0.13) disagreed with the sonic log (0.045 at 6-m

averaging) by up to a factor of four. With non-overlapping windows, all events, separate sensors and the domain effect, the discrepancy largely disappears. Adding the domain effect to the pink medium with the log value ($\sigma = 0.045$) gives median ratios of -0.33 (A) and $+0.42$ (B), within a factor of about 3 at both sensors. This agreement rests on one domain test at one sensor and should be read as consistency, not as a measurement.

Third, the log and the wavefield sample different volumes: one vertical line through intact rock outside the stimulated zone, and the paths through the zone being stimulated. Stimulation might raise the scattering strength by opening fluid-filled fractures, which lower V_s preferentially. This is a hypothesis. A stage-by-stage comparison is not possible, because 402 of the 416 events belong to stage 3 (the 14 stage-2 events show no significant difference; Additional file 1: Table S8). No pre-stimulation waveforms exist for these paths. The V_s -dominated medium shows only that a V_s -weighted perturbation would produce similar late energy, and the fracture-zone medium shows that a few planar zones are not enough; neither shows that stimulated fractures produced the coda.

Implications

For microseismic practice, the result means that velocity models adequate for locating events at ~ 1 -ms travel-time precision leave most of the late wavefield unexplained, and that even fine deterministic layering explains its level but not its spatial incoherence. The late wavefield carries information on small-scale 3-D structure, but it must be compared statistically, because the realisation cannot be recovered from two sensors. For the GFI concept, the result is necessary but not sufficient. Strong small-scale 3-D heterogeneity is required, and a lognormal power-law medium with roughly the logged σ provides it. The present seismic data do not, however, distinguish it from media with a short correlation length. Decisive tests of spectral shape need records at more stations, measured sensor transfer functions, larger simulation domains, and simulations above 100 Hz, where the families differ most. The 4-

kHz FORGE records contain S energy above noise up to about 1.8 kHz (Additional file 1: Figs. S5, S6), but fixed-frequency tool resonances near 330, 480 and 920 Hz currently prevent interpreting that band as source or path energy.

Limitations

- Two sensors in one well, with unoriented horizontals and unreliable vertical channels; no calibration shots, perforation shots or independent sensors for 2022, so the sensor–borehole transfer function is not measured.
- Sensor positions, clock offsets and V_P trade off with the catalogue locations (~ 50 m, $\sim 8\%$).
- The stochastic media are isotropic, lognormal and elastic, with constant density; intrinsic attenuation, anisotropic fracture compliance and rough interfaces are not modelled.
- The domain-size effect on the stochastic coda was measured for one medium and one sensor; the correction of σ_{eff} is approximate.
- The common grid has two realisations per grid point, fewer than a full ensemble; realisation scatter limits σ_{eff} to about a factor of 1.8 for pink and short- a media and much more for large- a Gaussian media.
- The analysis band is 20–80 Hz, where the spectral families differ least.

Conclusions

- Media without structure below ~ 50 m (homogeneous, 50-m layering, 3-D gradients, VTI) under-predict the late S energy of 416 FORGE MEQs by one to two orders of magnitude when the moment tensor is fitted to the direct waves only. The deficit is robust to windows, source constraints, source duration, event timing and a bounded in-band resonance, and the simulated coda is converged in grid and boundary.

- Deterministic 5–10 m structure (6-m sonic-log layering, thin fracture zones) supplies much of the late energy, but keeps the coda of neighbouring events coherent, whereas the observed coda decorrelates.
- Stochastic 3-D media reproduce both descriptors, with an effective scattering strength $\sigma_{\text{eff}} = 0.05\text{--}0.13$ in the reference domain. A larger domain lowers this by a factor of about 1.7, towards the sonic-log value of 0.045.
- Within 20–80 Hz the waveforms do not distinguish power-law, exponential and Gaussian spectra; the sonic log is compatible with a power law but does not exclude a short correlation length.
- Deterministic comparison against unknown stochastic realisations favours smooth media, so medium-class selection should use wavefield statistics.

Supplementary information

Additional file 1: Figures S1–S12 and Tables S1–S16 (reciprocity validation, sonic log, earlier statistical and shape tests, bandwidth and multi-octave diagnostics, window and moment-tensor sensitivity, S-timing subsets, calibration bootstrap, source spectra, ringing tests, stage comparison, field statistics, numerical convergence, σ_{eff} curves, complete Φ and holdout tables, equal- σ truth tests, log-spectrum variants and per-event results for every medium and seed).

Availability of data and materials

The GES 2022 catalogue is public [10]. The 56-32 SEG-2 records were provided to the authors; their public release is being confirmed with the data owner. All code (FD solver, reciprocity, features, tests; random seeds are fixed in the scripts) and derived result tables are at https://github.com/maswiet/Paper2_PinkNoise_WaveformInversion [COMMIT HASH AND ARCHIVE DOI TO BE ADDED AT SUBMISSION].

554 **Competing interests**

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558 **Authors' contributions**

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